Requirement / Specification	Domain	Source Document	Category	Subcategory
The child rover shall successfully transmit temperature, time-stamped video data and photos to the ground station and mother rover.	Emergency and First Response	argos_pdd.pdf	Information Sharing & Awareness	System Status Visibility
The child rover shall successfully receive GPS navigation coordinates from the ground station and mother rover.	Emergency and First Response	argos_pdd.pdf	Control & Coordination	Multi-Agent Coordination
The child rover shall be able to receive commands from the mother rover.	Emergency and First Response	argos_pdd.pdf	Control & Coordination	Multi-Agent Coordination
The child shall be able to receive commands from the ground station.	Emergency and First Response	argos_pdd.pdf	Team Capabilities & Authority	System Capabilities
The ground station shall confirm if the child is within +- 5 m of the location of interest.	Emergency and First Response	argos_pdd.pdf	Information Sharing & Awareness	System Status Visibility
The mother rover shall be able to command either child to navigate to specified GPS coordinates in real-time (Note: way-points might be used).	Emergency and First Response	argos_pdd.pdf	Control & Coordination	Multi-Agent Coordination
The mother rover shall be able to receive commands from the ground station.	Emergency and First Response	argos_pdd.pdf	Team Capabilities & Authority	System Capabilities
The mother rover shall be able to send data to the ground station.	Emergency and First Response	argos_pdd.pdf	Information Sharing & Awareness	System Status Visibility
Upon loss of communication, the child rover shall return to its last known GPS location.	Emergency and First Response	argos_pdd.pdf	Safety & Risk Mitigation	Robot Safe Mode
The rover should send images and temperature data as they are captured in order to provide low-latency information about the flame front to the ground station team	Emergency and First Response	argos_pdd.pdf	Information Sharing & Awareness	System Status Visibility
The child rover shall successfully communicate with the ground station and the mother rover at the required distance of 250 meters in an area with densely packed obstacles with obstacles (0.25 trees/m2).	Emergency and First Response	argos_pdd.pdf	Team Capabilities & Authority	System Capabilities
The mother rover shall be able to command video feed on/off.	Emergency and First Response	argos_pdd.pdf	Control & Coordination	Robot System Configuration
The entire MAV system must be man-packable, adhering to strict size and weight limitations.	Emergency and First Response	katrina_journal_article.pdf	Team Capabilities & Authority	Human Operational Constraints
For safety and observability reasons all flights shall be conducted within Line of Sight (LOS) of the flight team.	Emergency and First Response	katrina_journal_article.pdf	Safety & Risk Mitigation	Safety State Preservation
The system must support dynamic in-flight redirects as flight plans regularly change once airborne.	Emergency and First Response	katrina_journal_article.pdf	Control & Coordination	Range of Control (Levels of Automation)

One team member must have some domain expertise to help guide and direct the team in the field.	Emergency and First Response	katrina_journal_article.pdf	Team Capabilities & Authority	Human Operational Constraints
Any professional-grade MAV system designed to operate in urban environments must withstand signal loss and interference.	Emergency and First Response	katrina_journal_article.pdf	Team Capabilities & Authority	System Capabilities
The Pilot must be responsible for the flight and flight worthiness of the platform.	Emergency and First Response	katrina_journal_article.pdf	Control & Coordination	Responsibility Delineation
The Mission Specialist must be responsible for the payload and gathering the data targeted during the mission	Emergency and First Response	katrina_journal_article.pdf	Control & Coordination	Responsibility Delineation
The Flight Director must be responsible for overall safety of the public, the team, and the robot (priority: 1. public safety, 2. team safety, 3. robot safety).	Emergency and First Response	katrina_journal_article.pdf	Control & Coordination	Responsibility Delineation
The Pilot and Flight Director should be well aware of civilian airspace regulations (FAA Private Pilot's Written Exam level).	Emergency and First Response	katrina_journal_article.pdf	Team Capabilities & Authority	Human Operational Constraints
The vehicle must be grounded in the case of any sort of anomaly (unexpected high turbulence or temporary loss of communications).	Emergency and First Response	katrina_journal_article.pdf	Safety & Risk Mitigation	Robot Safe Mode
The MAV should be immediately grounded if a bystander or a manned aircraft are spotted by the Flight Director or any memmber of the team.	Emergency and First Response	katrina_journal_article.pdf	Safety & Risk Mitigation	Robot Safe Mode
The DUC and the UAM Coordinator must be capable of establishing shared situation awareness and agreeing on shared goals.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Information Sharing & Awareness	System Status Visibility
The UAM Coordinator shall have partial authority to adjust the DUC's autonomy level, allocation tasks, and strategic perfomance settings (e.g. prioritsation rules, separation criteria).	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Control & Coordination	Range of Control (Levels of Automation)
AI-based assistant logic must be transparent and traceable; systems with opaque ("black-box") AI are not suitable as they hinder understanding and accountability.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Communication & Decision Support	Decision Transparency
The DUC must monitor the mental state of the UAM Coordinator and factors such as workload, situational awareness and stress using physiological sensors and eye tracking.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Team Performance Monitoring	Human Performance Monitoring
The DUC shall provide explanations for its behaviour, actions, and recommendations upon request.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Communication & Decision Support	Decision Transparency

The DUC shall identify poor and suboptimal strategies/actions/solutions proposed by the UAM Coordinator.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Communication & Decision Support	Decision Support
DUC should be able to provide indication of having acknowledged the UAM Coordinators’ instructions/intentions.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Information Sharing & Awareness	Human Status Visibility
The DUC shall not interfere when the UAM Coordinator is involved in other critical communications or actions.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Team Capabilities & Authority	System Capabilities
DUC should be able to call for/direct the UAM Coordinator’s attention to important information (e.g., attention guidance).	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Communication & Decision Support	Decision Support
DUC should be able to continuously monitor the U-space and traffic opera- tions, providing real-time updates and alerts to the UAM Coordinator.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Information Sharing & Awareness	System Status Visibility
DUC should be able to monitor the U-space by collecting real-time data from multiple sources, including data from aircraft and weather.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Information Sharing & Awareness	Robot Data Availability
DUC should be able to generate status reports on U-space operations, inci- dents, and performance metrics.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Information Sharing & Awareness	Robot Data Availability
DUC should be able to detect potential conflicts between UAS/UAM vehicles, such as near-collisions or airspace violations.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Information Sharing & Awareness	System Status Visibility
DUC should be able to perform simulations and scenario planning to antici- pate future traffic patterns and potential U-space capacity issues.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Communication & Decision Support	Decision Support
DUC should be able to retrieve and present information on the request of the UAM Coordinator.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Information Sharing & Awareness	System Status Visibility
DUC should be able to infer what information the UAM Coordinator needs and present it.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Communication & Decision Support	Decision Support
DUC should be able to correctly determine and understand what the UAM Coordinator is trying to understand, for which an explanation is needed.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Communication & Decision Support	Decision Transparency

DUC should be able to demonstrate the relevance of an explanation for a decision/action.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Communication & Decision Support	Decision Transparency
DUC should be able to determine an appropriate level of abstraction of an explanation according to the task, situation, trust, and expertise of the UAM Coordinator.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Communication & Decision Support	Decision Transparency
DUC should be able to explain how it derived an output.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Communication & Decision Support	Decision Transparency
DUC should be able to explain how it works.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Communication & Decision Support	Decision Transparency
DUC should be able to provide indication of having acknowledged the UAM Coordinators' instructions/intentions.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Information Sharing & Awareness	Human Status Visibility
DUC should be able to understand and generate human natural language.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Team Capabilities & Authority	System Capabilities
DUC should be able to communicate using different modalities, including voice, text, and graphics (e.g., highlight areas on the map).	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Communication & Decision Support	Interface Design
DUC should be able to automatically adapt the modality of interactions to end-user states, preferences, and situations	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Team Capabilities & Authority	System Capabilities
DUC should be able to recommend actions/solutions.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Communication & Decision Support	Decision Support
DUC should be able to decide and implement actions within its performance envelope.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Team Capabilities & Authority	System Capabilities
The DUC should allow the UAM Coordinator to adjust some of the authority limits and constraints in decision making and action implementation.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Control & Coordination	Robot System Configuration

DUC should be able to propose and justify optimised solutions, where appli- cable.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Communication & Decision Support	Decision Support
DUC should be able to propose alternative strategies/actions/solutions.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Communication & Decision Support	Decision Support
DUC should be able to solve problems with the UAM Coordinator following a checklist.	Aerospace and Aviation	978-3-031-93721-7_18-1.pdf	Communication & Decision Support	Decision Support
The system shall maximize responders', citizens', and victims' life safety.	Emergency and First Response	Human_roles...first_response.pdf	Safety & Risk Mitigation	Safety State Preservation
The system shall facilitate incident stabilization by containing and mitigating hazards.	Emergency and First Response	Human_roles...first_response.pdf	Safety & Risk Mitigation	Safety State Preservation
Information must be delivered at varying abstraction levels (0-4) based on the responder's role and decision-making needs.	Emergency and First Response	Human_roles...first_response.pdf	Communication & Decision Support	Interface Design
The UV Specialist shall prevent the robot from negatively impacting the broader response system.	Emergency and First Response	Human_roles...first_response.pdf	Communication & Decision Support	Responsibility Delineation
High-level commanders shall receive necessary reports (e.g., Hazard Reports) regardless of whether data was collected by a human or a robot.	Emergency and First Response	Human_roles...first_response.pdf	Information Sharing & Awareness	Robot Data Availability
Robots shall facilitate response planning and maintaining awareness.	Emergency and First Response	Human_roles...first_response.pdf	Communication & Decision Support	Decision Support
Robots shall remove responders from dangerous situations, and allow immediate site feedback prior to human responder entry.	Emergency and First Response	Human_roles...first_response.pdf	Safety & Risk Mitigation	Safety State Preservation
Direct Human Teammates must minimize communication errors, improve problem solving, and maintain local situational understanding in order to efficiently and effectively complete assigned tasks.	Emergency and First Response	Human_roles...first_response.pdf	Team Capabilities & Authority	Human Operational Constraints
Team Leaders must formulate tasks and maintain mission situational understanding.	Emergency and First Response	Human_roles...first_response.pdf	Control & Coordination	Responsibility Delineation

Division Chiefs must review robot-derived information to develop response plans.	Emergency and First Response	Human_roles...first_response.pdf	Team Capabilities & Authority	Human Operational Constraints
Logistics Technical Specialists must allocate resources based on robot general status and procure backups for equipment failure.	Emergency and First Response	Human_roles...first_response.pdf	Team Capabilities & Authority	Human Operational Constraints
Staging Area Managers must use Robot General Status and Location information for effective placement of personnel and equipment.	Emergency and First Response	Human_roles...first_response.pdf	Team Capabilities & Authority	Human Operational Constraints
The autonomous system shall be designed to reduce the pilot's overall workload	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Team Capabilities & Authority	System Capabilities
The system shall operate in a manner that does not increase the pilot's stress levels during flight operations	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Team Capabilities & Authority	System Capabilities
Support systems must address pilot incapacitation due to workload, fatigue, or health issues to ensure safety.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Safety & Risk Mitigation	Safety State Preservation
The system shall provide clear and unambiguous communication to the human operator.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Communication & Decision Support	Interface Design
The system shall detect and mitigate risks associated with pilot incapacitation due to health, fatigue, or workload	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Team Performance Monitoring	Human Performance Monitoring
The system shall provide a manual override mechanism to allow for immediate human control input	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Safety & Risk Mitigation	Override and Shut-Down Capabilities
The system shall provide a continuous information output stream to maintain the operator's situational awareness	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Information Sharing & Awareness	System Status Visibility
All communication channels between the human and the autonomous system shall comply with established aviation reliability standards	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Team Capabilities & Authority	System Capabilities
The system shall monitor and recognize significant increases in pilot mental workload or cognitive burden	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Team Performance Monitoring	Human Performance Monitoring
All system-to-human communications shall be presented in a comprehensible format that maintains accurate operator knowledge	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Communication & Decision Support	Interface Design
The HMI shall be designed to ensure the operator maintains full situational awareness during all phases of flight	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Communication & Decision Support	Interface Design
The human-machine interface (HMI) shall support bidirectional communication between air and ground systems.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Communication & Decision Support	Interface Design

The system shall incorporate features specifically designed to maintain the operator's level of attention.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Communication & Decision Support	Interface Design
The ground operator shall receive functional support from both air-based and ground-based autonomous systems.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Communication & Decision Support	Decision Support
The air and ground interfaces shall ensure effective bidirectional communication between all parties.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Communication & Decision Support	Interface Design
The air-based autonomous system shall coordinate with the ground side to ensure a safe return-to-land outcome.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Safety & Risk Mitigation	Robot Safe Mode
The system shall enable dynamic adaptation to changing operational situations to support teamwork.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Team Capabilities & Authority	System Capabilities
The system shall ensure that all autonomous actions are compatible with the current situational context.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Team Capabilities & Authority	System Capabilities
All operational procedures shall be validated and refined based on direct feedback from pilots and ATCOs.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Team Capabilities & Authority	Human Operational Constraints
The system shall provide the human operator with historical decision context to ensure sufficient awareness during control take-over.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Communication & Decision Support	Decision Transparency
The system framework shall support the operational transition from Dual Pilot Operations (DPO) to Reduced Crew Operations (RCO).	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Control & Coordination	Range of Control (Levels of Automation)
The system framework shall support the operational transition to Single Pilot Operations (SiPO).	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Control & Coordination	Range of Control (Levels of Automation)
Autonomy levels and authority must align with operational needs, balancing automation efficiency with oversight.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Control & Coordination	Range of Control (Levels of Automation)
Control transitions must consider driver readiness, awareness, and transition time.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Control & Coordination	Responsibility Delineation
Robots must anticipate or comprehend the actions or needs of the human partner.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Team Capabilities & Authority	System Capabilities
Eye tracking should identify what the pilot is observing to link attention to situational awareness.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Team Performance Monitoring	Human Performance Monitoring
Notifications and reminders should be optimized to increase the accuracy of the pilot's world view.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Team Capabilities & Authority	System Capabilities
System support must be tailored to human skill level and experience to minimize cognitive burden.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Team Capabilities & Authority	System Capabilities
Effective teaming relies on sharing knowledge to maintain transparency in situational awareness.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Information Sharing & Awareness	System Status Visibility

Mechanisms must support accurate interpretation of human behavior and recognition of undesirable or abnormal performance.	Aerospace and Aviation	978-3-032-12392-3_4.pdf	Team Performance Monitoring	Human Performance Monitoring
A Policy Management (PolMan) automation shall maintain a database to ensure compliance and identify violations.	Military and Security	AD1183655.pdf	Team Capabilities & Authority	System Capabilities
A task manager must determine user tasks and divide them into a hierarchy for balanced task loading.	Military and Security	AD1183655.pdf	Control & Coordination	Responsibility Delineation
Plan monitors should evaluate execution progress and utilize strategies (autonomous or human) for correcting problems.	Military and Security	AD1183655.pdf	Team Performance Monitoring	Other
Teaming technologies must enable autonomous systems to operate robustly in hazardous and chaotic environments.	Military and Security	AD1183655.pdf	Team Capabilities & Authority	System Capabilities
Interface design practices must support bi-directional understanding of intent.	Military and Security	AD1183655.pdf	Communication & Decision Support	Interface Design
Systems must support dynamic work distribution and mission collaboration.	Military and Security	AD1183655.pdf	Control & Coordination	Responsibility Delineation
Working agreements must dynamically set parameters and thresholds for task re-allocation.	Military and Security	AD1183655.pdf	Control & Coordination	Robot System Configuration
Standard operator interface design guidelines shall be identified for the initiation of autonomous systems.	Military and Security	AD1183655.pdf	Control & Coordination	System Initiation
Standard operator interface design guidelines shall be identified for the monitoring of autonomous systems.	Military and Security	AD1183655.pdf	Communication & Decision Support	Interface Design
Systems shall provide a bi-directional understanding of intent and human-autonomy mission collaboration.	Military and Security	AD1183655.pdf	Information Sharing & Awareness	System Status Visibility
Interface design shall support dynamic work distribution and human-autonomy mission success.	Military and Security	AD1183655.pdf	Communication & Decision Support	Interface Design
Teams should utilize parameters and thresholds for dynamic task re-allocation tailored to operational workload.	Military and Security	AD1183655.pdf	Control & Coordination	Range of Control (Levels of Automation)
User modelling should support continuous monitoring of team members' states (e.g., workload), skills, and performance.	Military and Security	AD1183655.pdf	Team Performance Monitoring	Human Performance Monitoring
The system shall use an ontology that allows flexible interactive communication.	Military and Security	AD1183655.pdf	Team Capabilities & Authority	System Capabilities
An intelligent agent should clearly indicate what function it is currently performing to avoid mode errors.	Military and Security	AD1183655.pdf	Information Sharing & Awareness	Robot Mode Change Notification
The system shall integrate information from separate sources into a single shared mental model.	Military and Security	AD1183655.pdf	Information Sharing & Awareness	System Status Visibility

Both teammates must be able to monitor each other's decisions and raise concerns before actions are executed.	Military and Security	AD1183655.pdf	Team Performance Monitoring	Other
Agents should provide a representation of their own "self-confidence" in planned decisions to aid operator trust.	Military and Security	AD1183655.pdf	Communication & Decision Support	Decision Transparency
The system shall maintain meaningful human control across all AI systems.	Military and Security	AD1183655.pdf	Control & Coordination	Range of Control (Levels of Automation)
Assistant systems should mitigate erroneous behavior by stepwise increasing intervention, from alerts to overrides.	Military and Security	AD1183655.pdf	Safety & Risk Mitigation	Override and Shut-Down Capabilities
Agents must model other participants' intentions and actions.	Military and Security	AD1183655.pdf	Team Capabilities & Authority	System Capabilities
Human and agent team members shall be mutually predictable.	Military and Security	AD1183655.pdf	Team Capabilities & Authority	Other
Autonomous agents must be easily directable by human operators.	Military and Security	AD1183655.pdf	Control & Coordination	Range of Control (Levels of Automation)
Agents must make status and intentions obvious to teammates.	Military and Security	AD1183655.pdf	Information Sharing & Awareness	System Status Visibility
Agents must be capable of engaging in goal negotiation.	Military and Security	AD1183655.pdf	Team Capabilities & Authority	System Capabilities
Agents must participate in managing attention.	Military and Security	AD1183655.pdf	Team Capabilities & Authority	System Capabilities
The system shall provide an easily accessible communication channel.	Military and Security	AD1183655.pdf	Communication & Decision Support	Interface Design
The intelligent agent shall indicate its current function.	Military and Security	AD1183655.pdf	Information Sharing & Awareness	Robot Mode Change Notification
The system shall integrate information from separate sources into a single shared mental model.	Military and Security	AD1183655.pdf	Information Sharing & Awareness	System Status Visibility
The Human-Machine Interface (HMI) shall provide a transparent representation of the machine's decision-making stages.	Military and Security	AD1183655.pdf	Communication & Decision Support	Decision Transparency
The autonomous system shall communicate its understanding of the environment to humans.	Military and Security	AD1183655.pdf	Information Sharing & Awareness	System Status Visibility
The autonomous system shall communicate temporal constraints to humans.	Military and Security	AD1183655.pdf	Information Sharing & Awareness	System Status Visibility
The system shall enable the operator to understand past agent actions.	Military and Security	AD1183655.pdf	Information Sharing & Awareness	System Status Visibility
The system shall enable the operator to understand current agent actions.	Military and Security	AD1183655.pdf	Information Sharing & Awareness	System Status Visibility

The system shall enable the operator to understand future agent actions.	Military and Security	AD1183655.pdf	Information Sharing & Awareness	Robot Data Availability
The system shall dynamically adjust decision-making authority based on mission context.	Military and Security	AD1183655.pdf	Control & Coordination	Range of Control (Levels of Automation)
Dynamic task allocation shall consider current operator workload.	Military and Security	AD1183655.pdf	Control & Coordination	Range of Control (Levels of Automation)
Human oversight shall provide risk judgment in uncertain conditions.	Military and Security	AD1183655.pdf	Team Capabilities & Authority	Human Operational Constraints
The human worker shall always have the final decision authority.	Military and Security	AD1183655.pdf	Control & Coordination	Responsibility Delineation
The assistant system shall monitor the environment and detect/analyze danger.	Military and Security	AD1183655.pdf	Safety & Risk Mitigation	Safety State Preservation
The system shall provide explainability as a core principle for hybrid team collaboration.	Military and Security	AD1183655.pdf	Communication & Decision Support	Decision Transparency
Agents should represent "self-confidence" in planned decisions to allow trust calibration.	Military and Security	AD1183655.pdf	Communication & Decision Support	Decision Transparency
The system shall sustain power for field operations until re-supply is available.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	System Capabilities
The system shall provide a power runtime indicator to manage power resources to effectively plan mission durations	Emergency and First Response	1096961.1096965.pdf	Information Sharing & Awareness	Robot Data Availability
The system shall provide near-range color video acuity for key tasks such as maneuvering	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	System Capabilities
The system shall provide far-range acuity video sufficient for key tasks such as maneuvering, object identification and detailed inspection.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	System Capabilities
The system shall provide a color video field of view suitable for real-time tasks.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	System Capabilities
The system shall be robust and combine enhanced canine/human capabilities without existing weaknesses.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	System Capabilities
The system shall maintain the ability to use video in confined spaces and for short-range object identification .	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	System Capabilities
The system shall project remote situational awareness into compromised or collapsed structures or convey other types of information down range within line of sight.	Emergency and First Response	1096961.1096965.pdf	Information Sharing & Awareness	System Status Visibility
The system shall provide beyond-line-of-sight range to project remote situational awareness into compromised or collapsed structures.	Emergency and First Response	1096961.1096965.pdf	Information Sharing & Awareness	System Status Visibility

The system shall be designed to minimize initial training time for operator proficiency.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	Human Operational Constraints
The system shall be designed to minimize annual proficiency training for certification maintenance.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	Human Operational Constraints
The system shall minimize the number of operators required for operation.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	System Capabilities
The system shall meet usability standards to allow for effective task performance.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	Human Operational Constraints
The operator console shall be viewable and usable in different lighting conditions.	Emergency and First Response	1096961.1096965.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall be operable by users wearing personal protective equipment such as gloves, helmets, and eye protection.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	Human Operational Constraints
The dashboard shall allow for the monitoring of general system health, including orientation, communication strength, and power level.	Emergency and First Response	1096961.1096965.pdf	Team Performance Monitoring	Other
Equipment weight shall allow for movement and storage using existing methods and tools.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	Human Operational Constraints
The system's Mean Time Between Failures (MTBF) shall exceed the duration of a standard deployment.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	System Capabilities
The system shall be self-sustaining for 72 hours without external re-supply.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	System Capabilities
Field maintenance duration shall be minimized.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	System Capabilities
Field maintenance shall be performable with a minimum of specialized tools.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	Human Operational Constraints
The system shall provide a power runtime indicator for mission planning.	Emergency and First Response	1096961.1096965.pdf	Information Sharing & Awareness	Robot Data Availability
The system shall maintain working time beyond basic mobility for its terrain type.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	System Capabilities
The robot shall be a mechanism with locomotion and sensing, allowing remote human interaction.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	System Capabilities
The robot shall specify its maximum force available to push.	Emergency and First Response	1096961.1096965.pdf	Team Capabilities & Authority	System Capabilities
The system shall provide auto-notification when it requires assistance.	Emergency and First Response	1096961.1096965.pdf	Information Sharing & Awareness	System Status Visibility

The system shall be located at a comfortable distance from the user's position according to the required level of interaction.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Safety & Risk Mitigation	Environmental Factors
Workstation elements shall be designed to align user inputs with corresponding system outputs in a manner reflecting natural human behavior.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Safety & Risk Mitigation	Environmental Factors
The workstation and robotic system shall provide functions that adapt to the user's preferred working methods.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Team Capabilities & Authority	System Capabilities
The design should avoid similar types of multiple robotic systems interacting with the user to prevent them from being seen as threatening.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Interface Design
The design should avoid similar colors of multiple robotic systems interacting with the user.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Interface Design
The design should avoid similar appearances of multiple robotic systems interacting with the user.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Team Capabilities & Authority	System Capabilities
The system shall provide measures to adapt its behavior to correspond with a user.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Team Capabilities & Authority	System Capabilities
The system shall provide measures to adapt its interaction patterns to correspond with a user.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Team Capabilities & Authority	System Capabilities
The robotic system shall adapt its behavior based on previous interactions and collaborative work with the user.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Team Capabilities & Authority	System Capabilities
The robotic system shall adapt its communication mode based on previous interactions and collaborative work with the user.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Team Capabilities & Authority	System Capabilities
The robot system behavior shall be designed to be consistent, coherent, and comprehensible to the user.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Team Capabilities & Authority	System Capabilities
Multiple robotic systems shall avoid similar behaviors, such as movements, tasks, or decisions, when interacting with a user.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Control & Coordination	Multi-Agent Coordination
The robotic system shall transfer objects to the user strictly within the comfortable reach zone.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Safety & Risk Mitigation	Environmental Factors

The robotic system shall be enabled to foresee the user's intentions for physical interaction.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Team Performance Monitoring	Human Performance Monitoring
The robotic system shall be able to deal with the user's intended actions in advance.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Team Capabilities & Authority	System Capabilities
The robotic system shall be enabled to disregard unintended user actions that might trigger a false positive response.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Team Capabilities & Authority	System Capabilities
Actions of both the user and the robotic system shall be planned to avoid conflicts like collisions or misunderstandings.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Safety & Risk Mitigation	Environmental Factors
User-centered approaches shall be adopted to design pleasant interaction patterns.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Team Capabilities & Authority	System Capabilities
User-centered approaches shall be adopted to design corresponding human-machine interfaces.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Interface Design
The system shall prevent communicating erroneous intent by utilizing social conventions, such as handing over a tool by its handle.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Other
Users and robotic systems shall be supported to share the same communication model, such as language.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Other
The system shall use vocabulary that is simple and easy for the user to understand.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Other
The system shall suggest adequate work breaks to the user.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Team Performance Monitoring	Human Performance Monitoring
Robotic system interaction requests shall not distract or interfere with the user's motor activities.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Safety & Risk Mitigation	Environmental Factors
Robotic system interaction requests shall not distract or interfere with the user's attention.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Interface Design
Robotic system interaction requests shall not distract or interfere with the user's comprehension.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Team Performance Monitoring	Human Performance Monitoring

The user shall be made intuitively and immediately aware of the robotic system's status.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Information Sharing & Awareness	System Status Visibility
The user shall be made intuitively and immediately aware of the robotic system's behavior, including movements and tasks.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Information Sharing & Awareness	System Status Visibility
The user shall be made intuitively and immediately aware of the robotic system's intentions when relevant.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Information Sharing & Awareness	System Status Visibility
The amount of information provided shall be personalized according to user interaction preferences.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Other
The form of information provided shall be personalized according to user interaction preferences.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Other
The content of information provided shall be personalized according to user interaction preferences.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Other
The communication mode for information shall be personalized according to user interaction preferences.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Other
The user shall be allowed to provide feedback to confirm a proposed action plan.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Information Sharing & Awareness	Human Status Visibility
The user shall be allowed to provide feedback to reject a proposed action plan.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Information Sharing & Awareness	Human Status Visibility
The system shall provide measures for the robot to explain its decisions to the user.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Decision Transparency
The user shall be informed about the functioning of specific safety measures in use.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Decision Transparency
The user shall be informed about the type of specific safety measures in use.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Decision Transparency
Interfaces shall be designed to support the user in unambiguously understanding the information.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Interface Design

Interfaces shall be designed to support the user in easily understanding information through appropriate modality, format, and timing.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Interface Design
The robotic system shall be able to communicate apology statements and acknowledge errors.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Other
Robot-to-user communication shall be simplified by avoiding unnecessary or overly complex information.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Other
Information received by the user must be clear and unambiguous, avoiding contradictions or delays.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Interface Design
The system shall provide multimodal communication channels in a redundant way.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Team Capabilities & Authority	System Capabilities
The system shall provide measures to communicate with the user that do not cause a loss of focus on the task.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Interface Design
The user shall be allowed to understand a forthcoming task in advance.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Information Sharing & Awareness	System Status Visibility
The user shall be enabled to intuitively understand the robotic system's intentions beforehand.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Information Sharing & Awareness	System Status Visibility
The user shall be enabled to intuitively understand the spatial occupancy of planned robot motions beforehand.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Information Sharing & Awareness	System Status Visibility
The system shall signal its target and interested workpieces so the user understands intentions beforehand.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Information Sharing & Awareness	System Status Visibility
User control over the robotic system shall be as natural, intuitive, and explicit as possible.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Control & Coordination	Robot System Configuration
Workstation and robotic systems shall adapt safety strategies based on user preferences.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Control & Coordination	Robot System Configuration
The robotic system shall be designed to value and properly employ the operator's expertise and skills.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Control & Coordination	Robot System Configuration

The user shall be allowed to provide real-time corrections to key arbitrary robotic system states.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Control & Coordination	Range of Control (Levels of Automation)
The user shall be allowed to provide real-time corrections if they disagree with autonomous system behavior.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Control & Coordination	Range of Control (Levels of Automation)
The user shall be allowed to set the robotic system's level of autonomy.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Control & Coordination	Range of Control (Levels of Automation)
The effectiveness and reliability of safety measures shall be demonstrated to the user before interaction starts.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Decision Transparency
The efficiency and reliability of robotic system elements shall be demonstrated to the user before interaction starts.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Information Sharing & Awareness	System Status Visibility
The robotic system shall be presented such that it is perceived as a useful, effective, and reliable workmate.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Other
Common language shall be used when presenting the robotic system to users.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Other
Human-like terminology shall be used when presenting the robotic system to users.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Other
Terminology that highlights cooperativeness shall be used when presenting the robotic system.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Communication & Decision Support	Other
Ensure the user’s agency (i.e., the capacity to take informed decisions and actions independently) when delegating decisions and tasks to the robotic system.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Control & Coordination	Responsibility Delineation
Ensure the user's sense of control when delegating decisions and tasks to the robotic system.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Control & Coordination	Responsibility Delineation
Ensure the user's responsibility over the work when delegating decisions and tasks to the robotic system.	Industrial and Manufacturing	1-s2.0-S0003687024000231-main.pdf	Control & Coordination	Responsibility Delineation
The system shall set trajectories in such a way human body parts will not be easily trapped between the robot systems and the elements of the workstation.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors

The system shall set safe virtual-plane-systems which limit the robot to work in a defined volume.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall set space limiting functions which limit the robot to work in a defined volume.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall set collaborative robot speed limits for quasi-static contact.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall signal the transition between collaborative operations and other kind of operations.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Information Sharing & Awareness	Robot Mode Change Notification
The system shall monitor robot systems performance.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Team Performance Monitoring	Other
The system shall set access routes for operators and material movement.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall minimize specific mechanical hazards related to collisions with human body parts.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall set trajectories in such a way human body parts will not be easily hit by the robot systems.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall set trajectories to minimize energy exchange during unexpected collisions.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall set collaborative robot speeds limit for transient contact.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall use a safety-rated monitored stop function.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Robot Safe Mode
The system shall limit moving masses.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall set trajectories such that potential parts falling will limit collision damages.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors

The system shall avoid HRIs which require the use of upper limbs for long time during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall avoid HRIs requiring elbows above shoulder level for long durations during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall avoid HRIs which require force peaks during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall avoid HRIs requiring continuous finger-tip grasping.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall avoid HRIs which require high frequency and similar movements of upper limbs during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall avoid HRIs which require to maintain the workstation elements far to the body during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall avoid HRIs which require a vertical displacement outside the range between hips and shoulders during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall avoid HRIs which require frequent body movements during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
Motion planning shall avoid HRIs which require an asymmetric posture of booth neck and trunk during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
Motion planning shall avoid HRIs which require unsupported trunk backward inclination or harsh flexion during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
Motion planning shall avoid HRIs which require neck extension or hash flexion during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
Motion planning shall avoid HRIs which require unsupported head backward inclination or harsh inclination during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
Motion planning shall avoid HRIs which require a convex spinal curvature (if sitting) during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors

Motion planning shall avoid HRIs which require awkward upper arm postures during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
Motion planning shall avoid HRIs which require raised shoulder during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
Motion planning shall avoid HRIs which require unsupported upper arm elevation during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
Motion planning shall avoid HRIs which require extreme elbow flexion/extension AND extreme forearm rotation during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
Motion planning shall avoid HRIs which require extreme wrist deviation during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
Motion planning shall avoid HRIs which require extreme knee flexion during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
Motion planning shall avoid HRIs which require knee not flexed in standing postures during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
Motion planning shall avoid HRIs which require not-neutral ankle position during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
Motion planning shall avoid HRIs which require kneeling or crouching during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
Motion planning shall avoid HRIs which require very high knee angle (if sitting) during the assembly.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Safety & Risk Mitigation	Environmental Factors
The system shall implement human-aware motion planning.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Team Capabilities & Authority	System Capabilities
The system shall inform operators about the robot speed.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Information Sharing & Awareness	System Status Visibility
The system shall avoid misalignment in the use of production resources.	Industrial and Manufacturing	1-s2.0-S2212827120308350-main.pdf	Control & Coordination	Multi-Agent Coordination

Every component of the architecture that is causally involved in the robot’s reasoning and action planning must be inspectable.	Social and Service Robotics	frai-05-866920.pdf	Team Capabilities & Authority	Other
Social robots need to provide reasons for their behavior in a way understandable to naive human users.	Social and Service Robotics	frai-05-866920.pdf	Communication & Decision Support	Decision Transparency
Users shall be able to request an explanation from the robot at any time.	Social and Service Robotics	frai-05-866920.pdf	Communication & Decision Support	Decision Transparency
The robot shall be able to produce a self-explanation of its own behavior online during a running human-robot interaction.	Social and Service Robotics	frai-05-866920.pdf	Communication & Decision Support	Decision Transparency
The semantics of information exchange between different components should be well-defined in terms of their relevance to social interaction.	Social and Service Robotics	frai-05-866920.pdf	Team Capabilities & Authority	Other
Information exchange semantics should be reflected in the design of inter-component communication interfaces.	Social and Service Robotics	frai-05-866920.pdf	Team Capabilities & Authority	Other
The architecture must support empirically validated models for generating understandable and desirable explanations for socio-interactive behavior.	Social and Service Robotics	frai-05-866920.pdf	Team Capabilities & Authority	Other
The architecture must define interfaces for delivering information relevant for behavior explanations.	Social and Service Robotics	frai-05-866920.pdf	Team Capabilities & Authority	Other
The explanation generation models should be interpretable or explainable themselves.	Social and Service Robotics	frai-05-866920.pdf	Team Capabilities & Authority	Other
Information necessary for explaining a behavior are generated and updated incrementally within the architecture, parallel to behavior generation.	Social and Service Robotics	frai-05-866920.pdf	Team Capabilities & Authority	Other
The architecture should be capable of delivering explanations according to the needs of the user.	Social and Service Robotics	frai-05-866920.pdf	Communication & Decision Support	Decision Transparency
The architecture should be capable of understanding verbal requests from the user and dynamically adapting the explanation strategy.	Social and Service Robotics	frai-05-866920.pdf	Team Capabilities & Authority	System Capabilities
The architecture should support flexible dialog in order to ensure that the user requests can be served seamlessly, also within an ongoing conversation.	Social and Service Robotics	frai-05-866920.pdf	Team Capabilities & Authority	System Capabilities
A robot’s behavior should either be readily interpretable to the user, or it should be potentially explainable to the user.	Social and Service Robotics	frai-05-866920.pdf	Communication & Decision Support	Decision Transparency
The robot itself shall be able to produce a self-explanation of its own behavior, online and during a running human-robot interaction.	Social and Service Robotics	frai-05-866920.pdf	Communication & Decision Support	Decision Transparency

At least two distinct accesses to the mode switch shall be set up(e.g., a mechanical button reachable when the operator is outside the tractor and an access on the control panel)	Industrial and Manufacturing	safety-11-00024.pdf	Safety & Risk Mitigation	Override and Shut-Down Capabilities
The emergency stop should be clearly marked and easily accessible.	Industrial and Manufacturing	safety-11-00024.pdf	Safety & Risk Mitigation	Robot Safe Mode
The emergency stop shall only be reset by a deliberate manual action that does not cause a restart.	Industrial and Manufacturing	safety-11-00024.pdf	Control & Coordination	System Initiation
The emergency stop shall only permit a restart to occur.	Industrial and Manufacturing	safety-11-00024.pdf	Control & Coordination	System Initiation
The following information shall be provided: maximum response time for the emergency stop.	Industrial and Manufacturing	safety-11-00024.pdf	Information Sharing & Awareness	System Status Visibility
The following information shall be provided: maximum stopping time for the emergency stop.	Industrial and Manufacturing	safety-11-00024.pdf	Information Sharing & Awareness	System Status Visibility
Specific limitations on maximum driving and turning speeds for risky weather in autonomous mode shall be set up.	Industrial and Manufacturing	safety-11-00024.pdf	Control & Coordination	Robot System Configuration
Quality assessment is expected to start once the autonomous mode starts.	Industrial and Manufacturing	safety-11-00024.pdf	Team Performance Monitoring	Other
The system should have instant feedback on whether the series of input images is valid for autonomous navigation.	Industrial and Manufacturing	safety-11-00024.pdf	Communication & Decision Support	Decision Support
If input images are not valid, disable the autonomous mode	Industrial and Manufacturing	safety-11-00024.pdf	Safety & Risk Mitigation	Robot Safe Mode
Provide suggested control parameters according to the ground condition.	Industrial and Manufacturing	safety-11-00024.pdf	Communication & Decision Support	Decision Support
Robot screens should complement speech to convey complex information.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Interface Design
Screens may be used to complement and augment verbal communication.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Interface Design
Human–screen interaction modality includes touch.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Interface Design
Robot–screen interaction modality includes looking.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Interface Design
Robot–screen interaction modality includes pointing.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Interface Design
The robot reacts after interaction by updating visuals.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Interface Design
The robot reacts after interaction by making sound.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Interface Design

The robot reacts after interaction by speaking.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Interface Design
The system shall present related information at once.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Interface Design
Information shall be displayed simultaneously rather than sequentially.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Interface Design
The system shall present information during speech.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Interface Design
Visual information shall be presented while the robot is speaking.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Interface Design
The system shall avoid robot self-interaction.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Interface Design
Avoid additional robot gestures that could be distracting.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Interface Design
The system shall confirm user input.	Social and Service Robotics	3757279.3785557.pdf	Information Sharing & Awareness	Human Status Visibility
The robot should provide confirmation after users interact with the screen.	Social and Service Robotics	3757279.3785557.pdf	Information Sharing & Awareness	Human Status Visibility
Confirmation should be provided in non-visual forms such as brief verbal acknowledgments.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Interface Design
Confirmation should be provided via nodding.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Interface Design
The robot may indicate what it could do next as follow-up guidance after user interaction.	Social and Service Robotics	3757279.3785557.pdf	Communication & Decision Support	Decision Support
The drone shall autonomously deliver a message to a worker.	Industrial and Manufacturing	3757279.3785563.pdf	Control & Coordination	Multi-Agent Coordination
The system shall adopt a multi-modal approach integrating sound, structured voice advisories, and physical presence to establish authority and ensure compliance.	Industrial and Manufacturing	3757279.3785563.pdf	Communication & Decision Support	Interface Design
The system shall use sound and movement to signal intent.	Industrial and Manufacturing	3757279.3785563.pdf	Information Sharing & Awareness	System Status Visibility
The system shall feature a voice persona that balances authority with clarity.	Industrial and Manufacturing	3757279.3785563.pdf	Communication & Decision Support	Interface Design
The system shall include interaction flows designed to manage user confusion and non-compliance.	Industrial and Manufacturing	3757279.3785563.pdf	Team Capabilities & Authority	System Capabilities
Voice advisories from drones must be intelligible and socially acceptable.	Industrial and Manufacturing	3757279.3785563.pdf	Communication & Decision Support	Interface Design

The system must ensure accountability through logging.	Industrial and Manufacturing	3757279.3785563.pdf	Information Sharing & Awareness	Robot Data Availability
The system shall provide clear voice instructions without adding to sensory overload.	Industrial and Manufacturing	3757279.3785563.pdf	Communication & Decision Support	Interface Design
The drone shall approach users with a gentle trajectory.	Industrial and Manufacturing	3757279.3785563.pdf	Team Capabilities & Authority	System Capabilities
The drone shall approach users with an audible chime.	Industrial and Manufacturing	3757279.3785563.pdf	Communication & Decision Support	Interface Design
The system shall deliver structured, predictable messages to reduce cognitive load.	Industrial and Manufacturing	3757279.3785563.pdf	Communication & Decision Support	Interface Design
The system shall use closed-loop confirmation by requesting a "thumbs up" gesture.	Industrial and Manufacturing	3757279.3785563.pdf	Information Sharing & Awareness	Human Status Visibility
The system shall interpret a problem signal (e.g., "thumbs down") as a request for help.	Industrial and Manufacturing	3757279.3785563.pdf	Information Sharing & Awareness	Human Status Visibility
The drone shall instruct the user to contact a support agent when help is requested.	Industrial and Manufacturing	3757279.3785563.pdf	Communication & Decision Support	Decision Support
The system shall match response intensity to situation severity (Tiered Escalation logic).	Industrial and Manufacturing	3757279.3785563.pdf	Team Capabilities & Authority	System Capabilities
The drone shall issue a firm command using a sharp, urgent auditory alert for critical violations.	Industrial and Manufacturing	3757279.3785563.pdf	Communication & Decision Support	Interface Design
The system shall alert a human Safety Officer for live voice intervention if automated commands are ignored.	Industrial and Manufacturing	3757279.3785563.pdf	Control & Coordination	Multi-Agent Coordination
The drone should make sound during its approach to signal presence and intent.	Industrial and Manufacturing	3757279.3785563.pdf	Information Sharing & Awareness	System Status Visibility
Approach sounds should be noticeable but not alarming.	Industrial and Manufacturing	3757279.3785563.pdf	Information Sharing & Awareness	System Status Visibility
The voice persona tone must be professional.	Industrial and Manufacturing	3757279.3785563.pdf	Communication & Decision Support	Other
The voice persona must avoid sounding "too happy or chirpy."	Industrial and Manufacturing	3757279.3785563.pdf	Communication & Decision Support	Other
The system shall trigger escalation if the user fails to provide a directive acknowledgement command.	Industrial and Manufacturing	3757279.3785563.pdf	Team Capabilities & Authority	System Capabilities
The system shall instruct lost or stuck personnel to "stay put" until help arrives.	Industrial and Manufacturing	3757279.3785563.pdf	Communication & Decision Support	Decision Support
The system must use distinct sounds to indicate different levels of criticality.	Industrial and Manufacturing	3757279.3785563.pdf	Communication & Decision Support	Interface Design
The drone shall fly backward during an escort to maintain constant visual contact.	Industrial and Manufacturing	3757279.3785563.pdf	Team Capabilities & Authority	System Capabilities

Safety instructions for critical violations must be clear, direct commands to leave immediately.	Industrial and Manufacturing	3757279.3785563.pdf	Communication & Decision Support	Interface Design
The system shall allow users to issue verbal commands to change the robot's position in real time.	Healthcare and Assistive Robotics	3757279.3785634.pdf	Control & Coordination	Robot System Configuration
The system shall provide clear communication of the robot's intent.	Healthcare and Assistive Robotics	3757279.3785634.pdf	Information Sharing & Awareness	System Status Visibility
The system shall provide clear communication of the robot's state.	Healthcare and Assistive Robotics	3757279.3785634.pdf	Information Sharing & Awareness	System Status Visibility
The system shall allow users to issue verbal commands to change the robot's velocity in real time.	Healthcare and Assistive Robotics	3757279.3785634.pdf	Control & Coordination	Robot System Configuration
The system shall allow users to issue verbal commands to change the robot's force in real time.	Healthcare and Assistive Robotics	3757279.3785634.pdf	Control & Coordination	Robot System Configuration
The system shall provide verbal feedback in response to the user.	Healthcare and Assistive Robotics	3757279.3785634.pdf	Communication & Decision Support	Interface Design
The verbal feedback shall be either an assurance of the desired change or a clarifying question.	Healthcare and Assistive Robotics	3757279.3785634.pdf	Communication & Decision Support	Interface Design
Clear communication of a robot’s intent and state is required.	Healthcare and Assistive Robotics	3757279.3785634.pdf	Information Sharing & Awareness	System Status Visibility
The framework must couple user trajectory commands with real-time, transparent verbal feedback.	Healthcare and Assistive Robotics	3757279.3785634.pdf	Communication & Decision Support	Decision Transparency
The system shall proactively ask a clarifying question when a user utterance is unclear.	Healthcare and Assistive Robotics	3757279.3785634.pdf	Communication & Decision Support	Decision Support
Clarifying questions shall be open-ended and avoid suggesting specific options.	Healthcare and Assistive Robotics	3757279.3785634.pdf	Communication & Decision Support	Interface Design
The contextual reminder system shall support the specification of typical household locations.	Social and Service Robotics	3757279.3788654.pdf	Control & Coordination	Robot System Configuration
The contextual reminder system shall support the identification of relevant family members.	Social and Service Robotics	3757279.3788654.pdf	Control & Coordination	Robot System Configuration

The robot system must be reliable, safe, and human-aligned.	Social and Service Robotics	3757279.3788654.pdf	Team Capabilities & Authority	System Capabilities
The mobile robot shall serve as the primary user interface.	Social and Service Robotics	3757279.3788654.pdf	Communication & Decision Support	Interface Design
The system shall include a computer that functions as the controller.	Social and Service Robotics	3757279.3788654.pdf	Team Capabilities & Authority	System Capabilities
The scheduler continuously evaluates the robot’s next action.	Social and Service Robotics	3757279.3788654.pdf	Team Capabilities & Authority	System Capabilities
The system shall send images to a cloud-based VLM for high-level scene understanding when relevant.	Social and Service Robotics	3757279.3788654.pdf	Team Capabilities & Authority	System Capabilities
The system shall provide a button to configure privacy mode.	Social and Service Robotics	3757279.3788654.pdf	Information Sharing & Awareness	Robot Mode Change Notification
The system shall enter privacy mode by default after the final reminder of the day.	Social and Service Robotics	3757279.3788654.pdf	Information Sharing & Awareness	Robot Mode Change Notification
The system shall exit privacy mode before the first scheduled task of the following day.	Social and Service Robotics	3757279.3788654.pdf	Information Sharing & Awareness	Robot Mode Change Notification
The robot shall listen for the wake word "Hey, Temi".	Social and Service Robotics	3757279.3788654.pdf	Information Sharing & Awareness	Human Status Visibility
The wake word shall trigger the robot to begin listening for user speech.	Social and Service Robotics	3757279.3788654.pdf	Information Sharing & Awareness	Human Status Visibility
The system shall involve parents in the interaction loop to leverage family authority structures.	Social and Service Robotics	3757279.3788654.pdf	Team Capabilities & Authority	Human Operational Constraints
The robot shall be endowed with help-seeking capabilities for recovery from failures.	Social and Service Robotics	3757279.3788654.pdf	Safety & Risk Mitigation	Failure Recovery
The system shall help users practice cognitive strategies with activities to minimize the impact of MCI on daily life.	Healthcare and Assistive Robotics	10333869.pdf	Team Capabilities & Authority	System Capabilities
The system shall consist of a social robot coupled with a tablet display to support multimodal communication and accessibility.	Healthcare and Assistive Robotics	10333869.pdf	Control & Coordination	Robot System Configuration
The robot intervention shall support goal setting.	Healthcare and Assistive Robotics	10333869.pdf	Information Sharing & Awareness	Human Status Visibility
The robot intervention shall support content personalization.	Healthcare and Assistive Robotics	10333869.pdf	Team Capabilities & Authority	System Capabilities
The robot intervention shall provide encouragement for real-world transfer.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Other

The robot intervention shall provide ways to longitudinally maintain engagement.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Other
Translate clinic-based cognitive interventions to robot-delivered, home-based interventions.	Healthcare and Assistive Robotics	10333869.pdf	Team Capabilities & Authority	System Capabilities
The robot shall validate the user's proposed solution.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Decision Support
The robot shall modify training content over time to avoid monotony and predictability.	Healthcare and Assistive Robotics	10333869.pdf	Team Capabilities & Authority	System Capabilities
The robot shall ask users about their concerns to personalize intervention content.	Healthcare and Assistive Robotics	10333869.pdf	Information Sharing & Awareness	Human Status Visibility
The robot shall allow users to select which cognitive strategies they want to practice.	Healthcare and Assistive Robotics	10333869.pdf	Control & Coordination	Robot System Configuration
The robot shall help users learn and practice content using examples from their daily lives.	Healthcare and Assistive Robotics	10333869.pdf	Team Capabilities & Authority	System Capabilities
The robot shall help users recognize actionable steps in daily life to support their goals.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Decision Support
The robot shall ask users to reflect on their experiences with intervention strategies.	Healthcare and Assistive Robotics	10333869.pdf	Information Sharing & Awareness	Human Status Visibility
The robot shall reward users based on effort rather than task performance.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Other
The robot shall track and show the user's progress toward goals over time.	Healthcare and Assistive Robotics	10333869.pdf	Team Performance Monitoring	Human Performance Monitoring
The difficulty of each activity shall be configured to a person's cognitive abilities.	Healthcare and Assistive Robotics	10333869.pdf	Control & Coordination	Robot System Configuration
Establishing goals is essential for improving motivation to engage in an intervention.	Healthcare and Assistive Robotics	10333869.pdf	Control & Coordination	Other

The robot shall track and show the user's progress toward goals over time.	Healthcare and Assistive Robotics	10333869.pdf	Team Performance Monitoring	Human Performance Monitoring
The system shall allow the user to take breaks at their own discretion.	Healthcare and Assistive Robotics	10333869.pdf	Team Capabilities & Authority	Human Operational Constraints
The robot shall check in with the user to determine if a break is needed.	Healthcare and Assistive Robotics	10333869.pdf	Team Performance Monitoring	Human Performance Monitoring
The robot shall cue the user to continue the session if distraction is detected.	Healthcare and Assistive Robotics	10333869.pdf	Team Performance Monitoring	Human Performance Monitoring
The robot shall provide encouragement to the user, specifically during moments of frustration.	Healthcare and Assistive Robotics	10333869.pdf	Team Performance Monitoring	Human Performance Monitoring
The tablet interface shall have buttons spaced apart to prevent accidental taps.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Interface Design
The robot shall use familiar communication modalities, including speech.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Interface Design
The robot shall repeat information upon request or for review.	Healthcare and Assistive Robotics	10333869.pdf	Information Sharing & Awareness	Robot Data Availability
The robot shall minimize internal distractions (e.g., unnecessary lights or movements) during information delivery.	Healthcare and Assistive Robotics	10333869.pdf	Information Sharing & Awareness	Robot Data Availability
Visuals must be big.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Interface Design
Visuals must be high contrast.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Interface Design
Visuals must be clear.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Interface Design
Visuals must be not too busy.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Interface Design

Speech must be clear.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Interface Design
Speech must be understandable.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Interface Design
The system shall allow users to adjust speech attributes, such as volume and speed, to their needs.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Interface Design
The robot's communication shall be concise to help the user maintain focus.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Interface Design
The robot shall use physical cues, such as gestures, to sustain user engagement.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Interface Design
The robot shall repeat information upon request or for review.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Decision Support
The robot shall minimize internal distractions (e.g., unnecessary lights or movements) during information delivery.	Healthcare and Assistive Robotics	10333869.pdf	Information Sharing & Awareness	System Status Visibility
The robot's physical behavior shall be minimal when providing critical information.	Healthcare and Assistive Robotics	10333869.pdf	Information Sharing & Awareness	System Status Visibility
The robot shall offer breaks to the user following long or challenging tasks.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Decision Support
The robot shall use physical cues, such as gestures, to sustain user engagement.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Interface Design
The robot shall utilize real-world items, such as grocery lists, within activities.	Healthcare and Assistive Robotics	10333869.pdf	Team Capabilities & Authority	System Capabilities
The robot shall reward users for consistent engagement over a specified number of sessions.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Other
The robot shall reward users for maintaining an interaction "streak."	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Other

The robot shall reward users for attempting new intervention content.	Healthcare and Assistive Robotics	10333869.pdf	Communication & Decision Support	Other
The robot shall obtain and utilize implicit feedback, such as task performance.	Healthcare and Assistive Robotics	10333869.pdf	Team Performance Monitoring	Human Performance Monitoring
The robot shall ask users for explicit feedback regarding their preferences.	Healthcare and Assistive Robotics	10333869.pdf	Information Sharing & Awareness	Human Status Visibility
The robot shall automatically adjust the difficulty of activities based on user performance.	Healthcare and Assistive Robotics	10333869.pdf	Team Capabilities & Authority	System Capabilities
The robot shall modify its communication modalities to suit the user's sensory or physical abilities.	Healthcare and Assistive Robotics	10333869.pdf	Team Capabilities & Authority	System Capabilities
The system shall collaboratively move an object to a goal pose.	Industrial and Manufacturing	2502.03346v1.pdf	Team Capabilities & Authority	System Capabilities
Collaboratively move an object to a goal pose.	Industrial and Manufacturing	2502.03346v1.pdf	Team Capabilities & Authority	System Capabilities
The agents shall observe the actions of one another.	Industrial and Manufacturing	2502.03346v1.pdf	Team Performance Monitoring	Other
The user and the robot hold opposite ends of an object (a wooden stick).	Industrial and Manufacturing	2502.03346v1.pdf	Control & Coordination	Multi-Agent Coordination
Facilitating effective communication among team members.	Healthcare and Assistive Robotics	2502.18688v1.pdf	Team Capabilities & Authority	System Capabilities
The system shall adapt to rapidly changing situations.	Healthcare and Assistive Robotics	2502.18688v1.pdf	Team Capabilities & Authority	System Capabilities
The system shall communicate supply recommendations to the team.	Healthcare and Assistive Robotics	2502.18688v1.pdf	Communication & Decision Support	Decision Support
The system shall be designed to reduce care delays.	Healthcare and Assistive Robotics	2502.18688v1.pdf	Team Capabilities & Authority	System Capabilities
The system shall indicate which errors occur and when they occur to provide decision-support.	Healthcare and Assistive Robotics	2502.18688v1.pdf	Communication & Decision Support	Decision Support

System communication must be clear to the user.	Healthcare and Assistive Robotics	2502.18688v1.pdf	Communication & Decision Support	Interface Design
System communication must be provided quickly to the user.	Healthcare and Assistive Robotics	2502.18688v1.pdf	Communication & Decision Support	Interface Design
System communication must be provided in an intuitive way.	Healthcare and Assistive Robotics	2502.18688v1.pdf	Communication & Decision Support	Interface Design
The system shall not interrupt human autonomy.	Healthcare and Assistive Robotics	2502.18688v1.pdf	Control & Coordination	Responsibility Delineation
The system shall only provide assistance when specifically prompted.	Healthcare and Assistive Robotics	2502.18688v1.pdf	Control & Coordination	Responsibility Delineation
The system shall assist healthcare workers accurately.	Healthcare and Assistive Robotics	2502.18688v1.pdf	Team Capabilities & Authority	System Capabilities
The system shall assist healthcare workers reliably.	Healthcare and Assistive Robotics	2502.18688v1.pdf	Team Capabilities & Authority	System Capabilities
The system shall assist workers passively while inside the patient's room.	Healthcare and Assistive Robotics	2502.18688v1.pdf	Control & Coordination	Responsibility Delineation
The system shall assist workers deliberately when outside the patient's room.	Healthcare and Assistive Robotics	2502.18688v1.pdf	Control & Coordination	Responsibility Delineation
Healthcare workers' autonomy must be preserved to allow them to override the robot when needed.	Healthcare and Assistive Robotics	2502.18688v1.pdf	Safety & Risk Mitigation	Override and Shut-Down Capabilities
The autonomous system shall have the ability to act independently to accomplish its task.	Military and Security	3610977.3635005.pdf	Team Capabilities & Authority	System Capabilities
The security robot shall address the growing complexities of safeguarding private and public spaces.	Military and Security	3610977.3635005.pdf	Team Capabilities & Authority	System Capabilities
Security robots shall gain public acceptance to facilitate their deployment in public spaces.	Military and Security	3610977.3635005.pdf	Team Capabilities & Authority	Other
The human operator shall control the robot to take action when they feel necessary.	Military and Security	3610977.3635005.pdf	Control & Coordination	Range of Control (Levels of Automation)

In low autonomy mode, the robot shall be incapable of completing actions independently.	Military and Security	3610977.3635005.pdf	Control & Coordination	Range of Control (Levels of Automation)
In high-risk, low-autonomy scenarios, the operator shall control the robot to intervene immediately.	Military and Security	3610977.3635005.pdf	Control & Coordination	Range of Control (Levels of Automation)
The semi-autonomous robot shall have the capability to request the operator to take over control.	Military and Security	3610977.3635005.pdf	Control & Coordination	Responsibility Delineation
The semi-autonomous robot shall continuously monitor its surroundings.	Military and Security	3610977.3635005.pdf	Information Sharing & Awareness	Robot Data Availability
The operator shall be able to take control of the semi-autonomous robot at any time.	Military and Security	3610977.3635005.pdf	Control & Coordination	Range of Control (Levels of Automation)
In high-risk, semi-autonomous scenarios, the robot shall request operator control to intervene immediately.	Military and Security	3610977.3635005.pdf	Control & Coordination	Range of Control (Levels of Automation)
The fully autonomous robot shall have the capability to detect threats on its own.	Military and Security	3610977.3635005.pdf	Team Capabilities & Authority	System Capabilities
The fully autonomous robot shall have the capability to take action when it deems them necessary.	Military and Security	3610977.3635005.pdf	Team Capabilities & Authority	System Capabilities
The fully autonomous robot shall continuously monitor its surroundings for any potential risks.	Military and Security	3610977.3635005.pdf	Safety & Risk Mitigation	Safety State Preservation
The fully autonomous robot shall determine when to act.	Military and Security	3610977.3635005.pdf	Team Capabilities & Authority	System Capabilities
The fully autonomous robot shall complete actions independently from a human operator.	Military and Security	3610977.3635005.pdf	Control & Coordination	Range of Control (Levels of Automation)
In high-risk, fully autonomous scenarios, the robot shall be able to intervene immediately upon threat detection.	Military and Security	3610977.3635005.pdf	Safety & Risk Mitigation	Robot Safe Mode